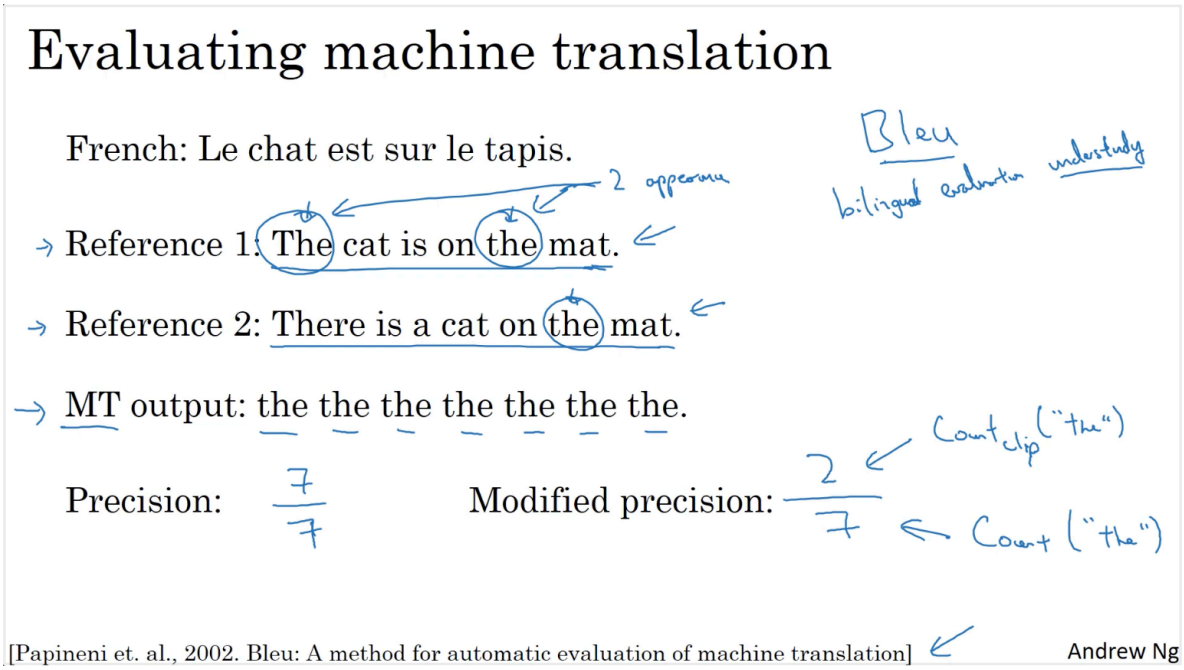
C5W3L06 Bleu Score (Optional)

Example: for French sentence you may have some english translations, but how to understand what's the best one?
Bleu is single number evaluation metric for language modeling, or machine translation systems, because we have multiple possible outputs

Note: If machine translation close to any of two references, it will have high Bleu score
For MT output we can calculate precision and modified precision (max between appearances in each reference)



Bleu score may be calculated on bigrams, Count_clip shows how frequent this bigram appeared in any of two references

Bleu score on bigrams

Example: Reference 1: The cat is on the mat.

Reference 2: There is a cat on the mat.

MT output: The cat the cat on the mat.

	Count	Count_{clip}
the cat	2	1
cat the	1	0
cat on	1	1
on the	1	1
the mat	1	1

$\frac{4}{6}$

[Papineni et. al., 2002. Bleu: A method for automatic evaluation of machine translation]

General case described

Bleu score on unigrams

Example: Reference 1: The cat is on the mat.

Reference 2: There is a cat on the mat.

MT output: The cat the cat on the mat. ($\hat{y}$)

$$p_1 = \frac{\sum_{\text{unigrams} \in \hat{y}} \text{Count}_{clip}(\text{unigram})}{\sum_{\text{unigram} \in \hat{y}} \text{Count}(\text{unigram})}$$

unigram

n-gram

$$p_n = \frac{\sum_{\text{n-grams} \in \hat{y}} \text{Count}_{clip}(\text{n-gram})}{\sum_{\text{n-grams} \in \hat{y}} \text{Count}(\text{n-gram})}$$

[Papineni et. al., 2002. Bleu: A method for automatic evaluation of machine translation] Andrew Ng

Actually Bleu score is computed using some n values (not only unigrams or bigrams like we have seen before above)

Bleu details

p_n = Bleu score on n-grams only

p_1, p_2, p_3, p_4

Combined Bleu score: $BP \exp\left(\frac{1}{4}\sum_{n=1}^{4} p_n\right)$

BP = brevity penalty

$$BP = \begin{cases} 1 & \text{if MT_output_length} > \text{reference_output_length} \\ \exp(1 - \text{MT_output_length}/\text{reference_output_length}) & \text{otherwise} \end{cases}$$

[Papineni et. al., 2002. Bleu: A method for automatic evaluation of machine translation]

Andrew Ng

BP is a special factor that forces output sequences to be not short